Physics-Grounded Data Augmentation Outperforms Model Adaptation for Cross-Dataset HAR Transfer

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April 2026

Motivation: Models Don't Travel Well

The Problem

Train a HAR classifier on UCI HAR $\rightarrow$ deploy on WISDM

Result: F1 = 0.085 (near random chance for 5 classes)

Why? The two datasets use:

- Different sensor placements (waist vs. front pocket)
- Different sampling rates (50 Hz vs. 20 Hz)
- Different signal processing (filtered vs. raw)

Domain shift is the culprit.
The question is: *which kind of domain shift, and can we fix it?*

Two Very Different Datasets

Property	UCI HAR	WISDM v1.1
Placement	Waist (fixed)	Front pocket (free)
Sampling rate	50 Hz	20 Hz
Filtering	Butterworth LP	None (raw)
Subjects	30	36
Training windows	8,355	—
Test windows	—	5,307
Units	m/s^2	m/s^2
Window size (aligned)	51×3 @ 20 Hz	

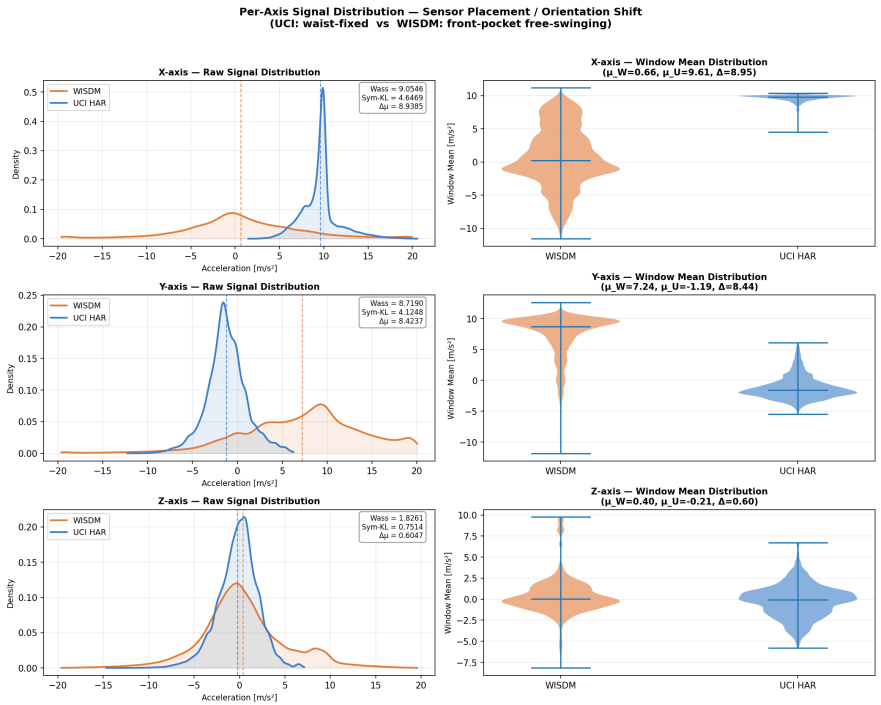
Same activity classes, but the sensor sees the world very differently.

Seven Sources of Domain Shift

Rank	Source	Magnitude
1	**Gravity axis projection**	$W_1 = 8.7\,\mathbf{m/s^2}$
2	Signal dynamic amplitude	3–5× difference
3	Noise temporal structure	AR(1) $\alpha \approx 0.986$ vs. white
4	Class prior distribution	TVD $= 0.36$
5	Gait spectral energy (0.8–2 Hz)	WISDM +40%
6	Inter-subject variability	$W_1^{\max} = 1.89\,\mathrm{m/s^2}$
7	Gravity magnitude attenuation	9.69 vs. 7.29 $\mathrm{m/s^2}$

Gravity axis mismatch completely dominates.

Dominant Shift: Gravity Axis Projection



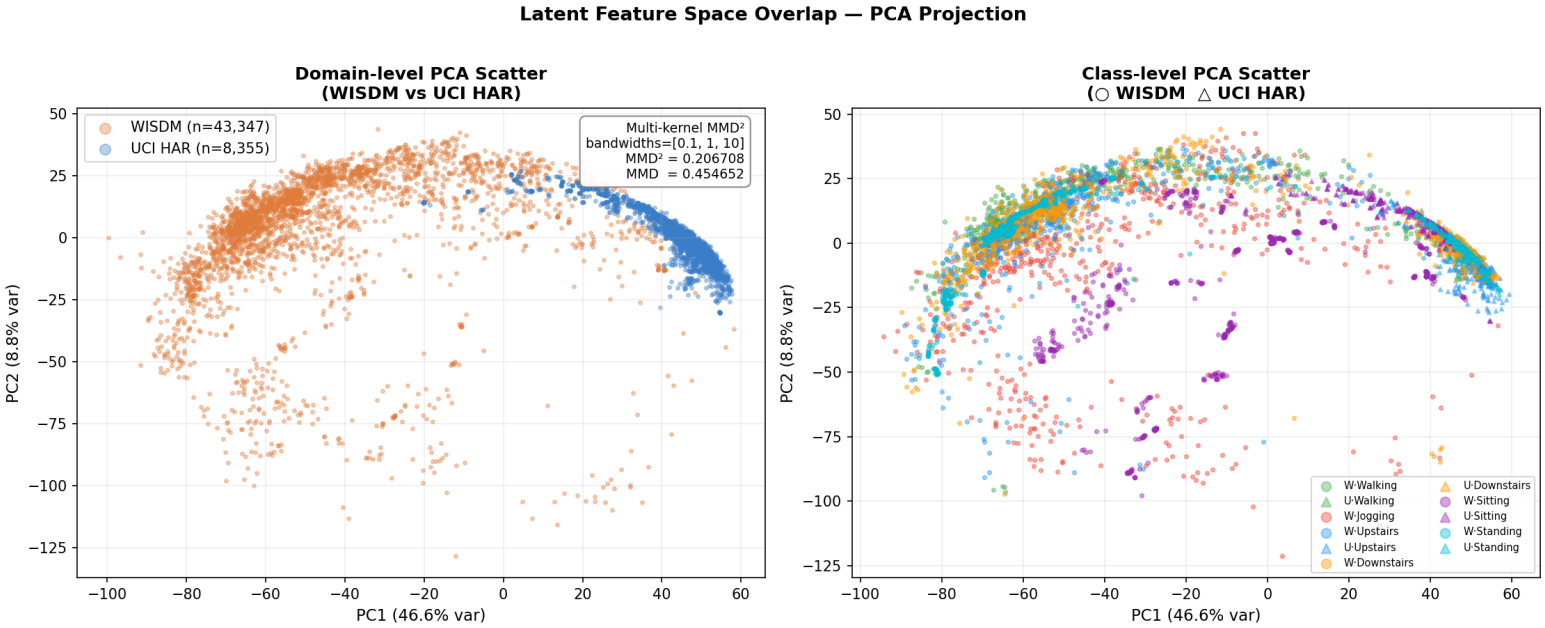
What's happening:

- Waist mount: gravity → **X-axis**
  Mean $= +8.58\,\mathrm{m/s^2}$
- Front pocket: gravity → **Y-axis**
  Mean $= +6.51\,\mathrm{m/s^2}$
- X-axis $W_1 = \mathbf{8.7}\,\mathrm{m/s^2}$

Sensor is physically rotated $\sim 90°$.

A model trained on UCI HAR learns "X has big DC." In WISDM, X is near zero-mean. This is why raw transfer fails.

How Bad Is the Gap in Feature Space?



Multi-kernel MMD = 0.830 Two datasets barely overlap. Raw transfer has no chance.

Our Approach: PhysicalDistortion

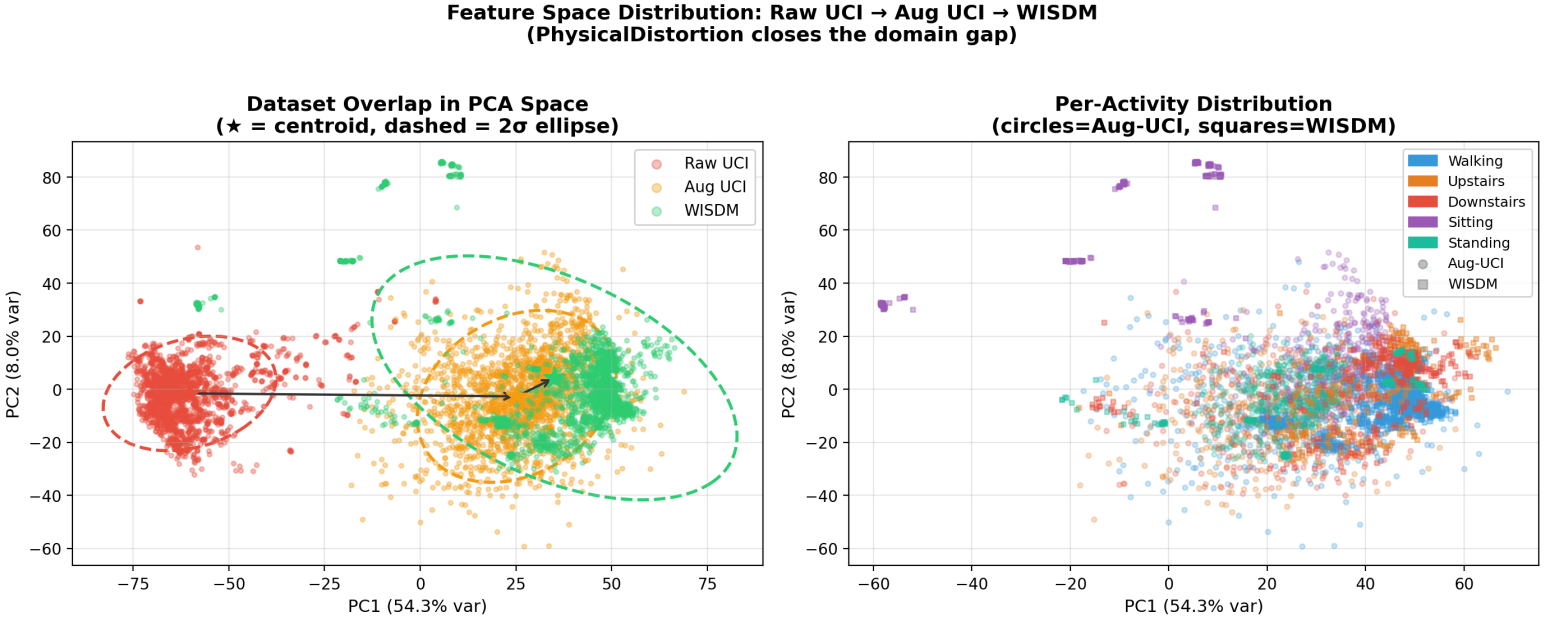
Instead of adapting the model, we transform the *data*.

Five-operator pipeline applied to UCI HAR training data:

1. **Orientation shift**: rotate 90° around Z + random wobble ±10°
2. **Gravity attenuation**: scale by $0.7523 = 7.29/9.69$
3. **Per-activity amplitude scaling**: empirical per-axis scale factors (not global)
4. **Gait spectral boost**: $\times$Uniform[1.2, 1.5] in 0.8–2 Hz band (locomotion only)
5. **AR(1) colored noise**: $\alpha = 0.9865$, initialized from stationary distribution

Each operator targets a specific, *measured* physical difference.
No target labels used. No model architecture changes.

After PhysicalDistortion: Distributions Align



MMD: 0.830 → 0.453 (45.4% reduction) Gap shrinks but doesn't close — explains remaining per-class failures.

Ablation: Four Conditions, Five Architectures

C0 Raw UCI HAR, no adaptation
C1 UCI HAR + PhysicalDistortion
C2 C1 + Test-Time Batch Norm (TTBN)
C3 C1 + Domain-Adversarial Training (DANN)

Model	C0 Raw	C1 Distort	C2 +TTBN	C3 +DANN
CNN1D	0.120	**0.723**	0.610	0.581
Transformer	0.046	**0.580**	0.580	0.327
FFN	0.069	**0.542**	0.425	0.401
TCN	0.127	**0.535**	0.451	0.418
BiGRU	0.062	**0.512**	0.512	0.365
Average	0.085	**0.578**	0.516	0.418

- C0 Raw: essentially random (6.8× improvement from C1)
- **C1 is best across every architecture**
- TTBN and DANN both hurt compared to C1 alone

Per-class Analysis: Honest Look

CNN1D, C1 condition:

Activity	F1
Walking	0.872
Sitting	0.966
Standing	0.820
Downstairs	0.513
Upstairs	0.445

Static activities: nearly perfect.

Upstairs is a real failure. Step patterns vary a lot across subjects and sensor wobble is hard to correct at window level.

Why TTBN and DANN Don't Help

After PhysicalDistortion, model-level adaptation makes things worse.

TTBN fails because:

- Re-estimates BN stats from test batches
- WISDM is class-imbalanced (Walking = 56% vs. UCI 20%, TVD = 0.36)
- Small test batches give noisy estimates that push stats in wrong direction

DANN fails because:

- Forces domain-invariant features
- But the shift is tied to activity itself (gravity projection differs per activity)
- Invariance destroys axis-specific information the classifier needs
- Training is also unstable; adversarial loss weight is hard to tune

Key insight: When shift is physical and correctable, correct it first. Model-level methods assume there's something left to adapt.

Lessons Learned

1. **Diagnose before adapting.** Understanding *why* distributions differ is more valuable than applying a generic adaptation method. Here, 90% of the shift traces to sensor placement geometry.
2. **Physical corrections are powerful.** Addressing root causes in data space outperformed two model-level baselines. The approach is also lightweight: no target labels, no new parameters.
3. **Model adaptation can make things worse.** TTBN and DANN both reduced performance when applied on top of good data augmentation. The combination is not always additive.
4. **Be honest about what's left.** Upstairs F1 $= 0.445$ is a failure. Inter-subject variability within WISDM (max $W_1 = 1.89\,\mathrm{m/s^2}$) is a real unsolved problem—our pipeline calibrates to dataset-level means, not individuals.

Conclusion

Main Finding

Physical differences between sensor setups drive cross-dataset HAR failure more than any feature-level mismatch. Correcting those physical differences in data space is more effective than adapting the model.

Numbers:

- Baseline (no adaptation): F1 = 0.085
- PhysicalDistortion: **F1 = 0.723** (6.0× for CNN1D; 6.8× avg)
- After HP search: **F1 = 0.762** (10,277-param CNN1D)
- TTBN/DANN on top: F1 drops to 0.610 / 0.581
- MMD: 0.830 → 0.453 (45% reduction)

Fix the physics first. Then think about the model.

Questions?

"Fix the physics first. Then think about the model."

Code and data available in the project repository.